

Employee Attrition Prediction

IBM HR Analytics Dataset — Mini Project Report

Classification modelling to identify employees at risk of leaving the organisation

1. Introduction

Employee attrition is a significant challenge for organisations, leading to increased recruitment costs, loss of institutional knowledge, and reduced productivity. This project applies supervised machine learning classification techniques to the IBM HR Analytics dataset to predict which employees are likely to leave.

Objectives

- Identify key factors that drive employee attrition
- Train and compare five classification models
- Handle class imbalance using SMOTE
- Evaluate models using precision, recall, F1, and ROC-AUC
- Provide actionable business recommendations

2. Dataset Overview

The IBM HR Analytics Employee Attrition dataset contains **1,470 employee records** with **35 features** covering demographics, job characteristics, satisfaction scores, and compensation details. The target variable is **Attrition** (Yes/No).

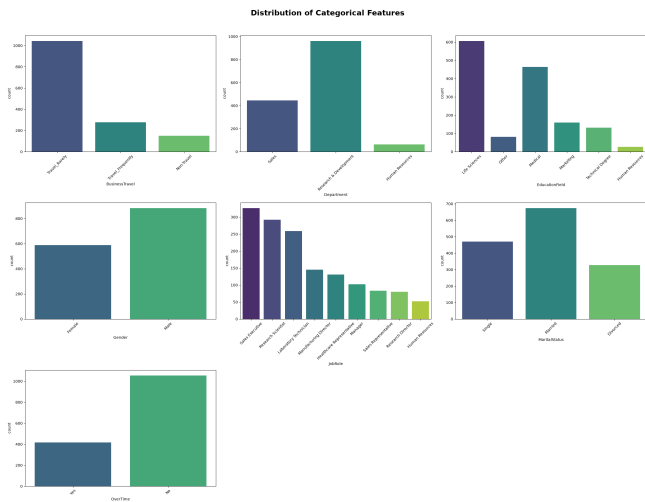
Attribute	Value
Total Records	1,470
Total Features	35 (31 after dropping constants)
Target Variable	Attrition (Yes / No)
Class Distribution	~84% No ~16% Yes
Missing Values	None
Duplicate Records	None

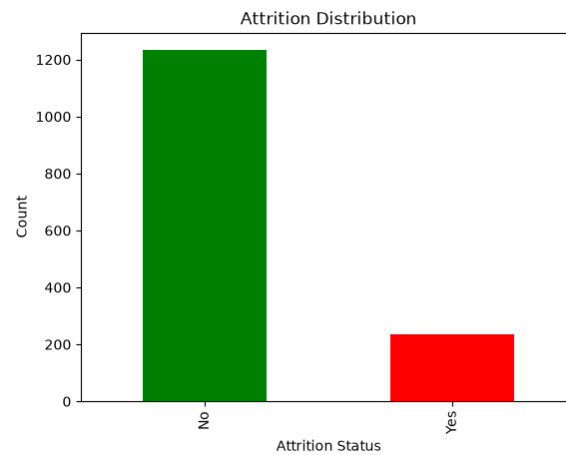
3. Exploratory Data Analysis

EDA was performed to understand the distribution of features, identify patterns, and uncover attrition drivers prior to modelling.

3.1 Attrition Distribution

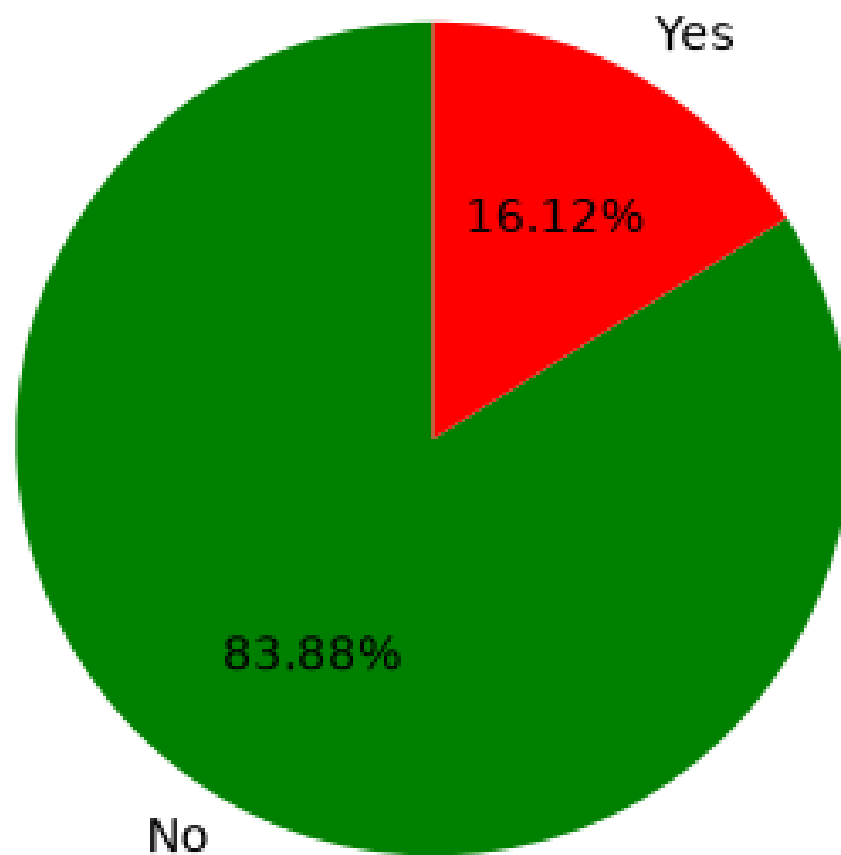
~16% of employees left. The dataset is significantly imbalanced — addressed using SMOTE.





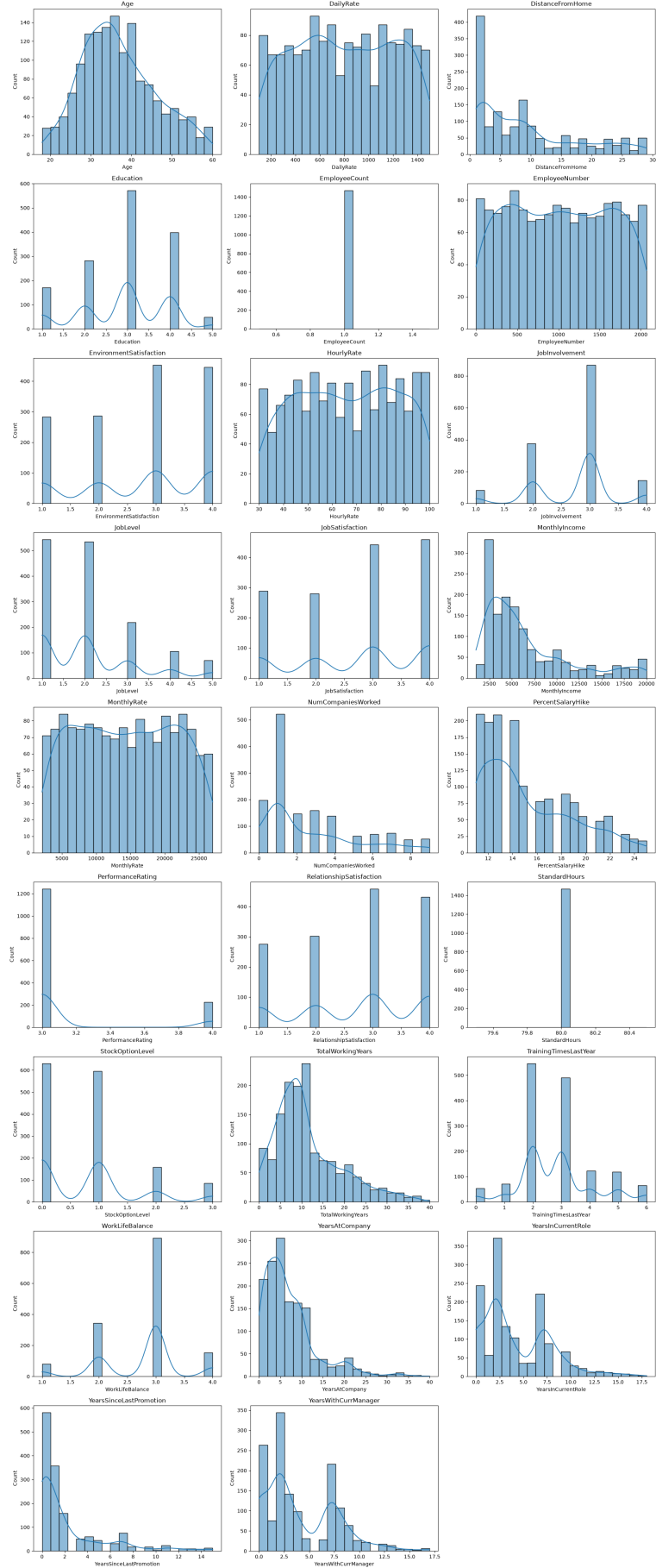
3.2 Categorical Feature Distributions

Employee Attrition Distribution



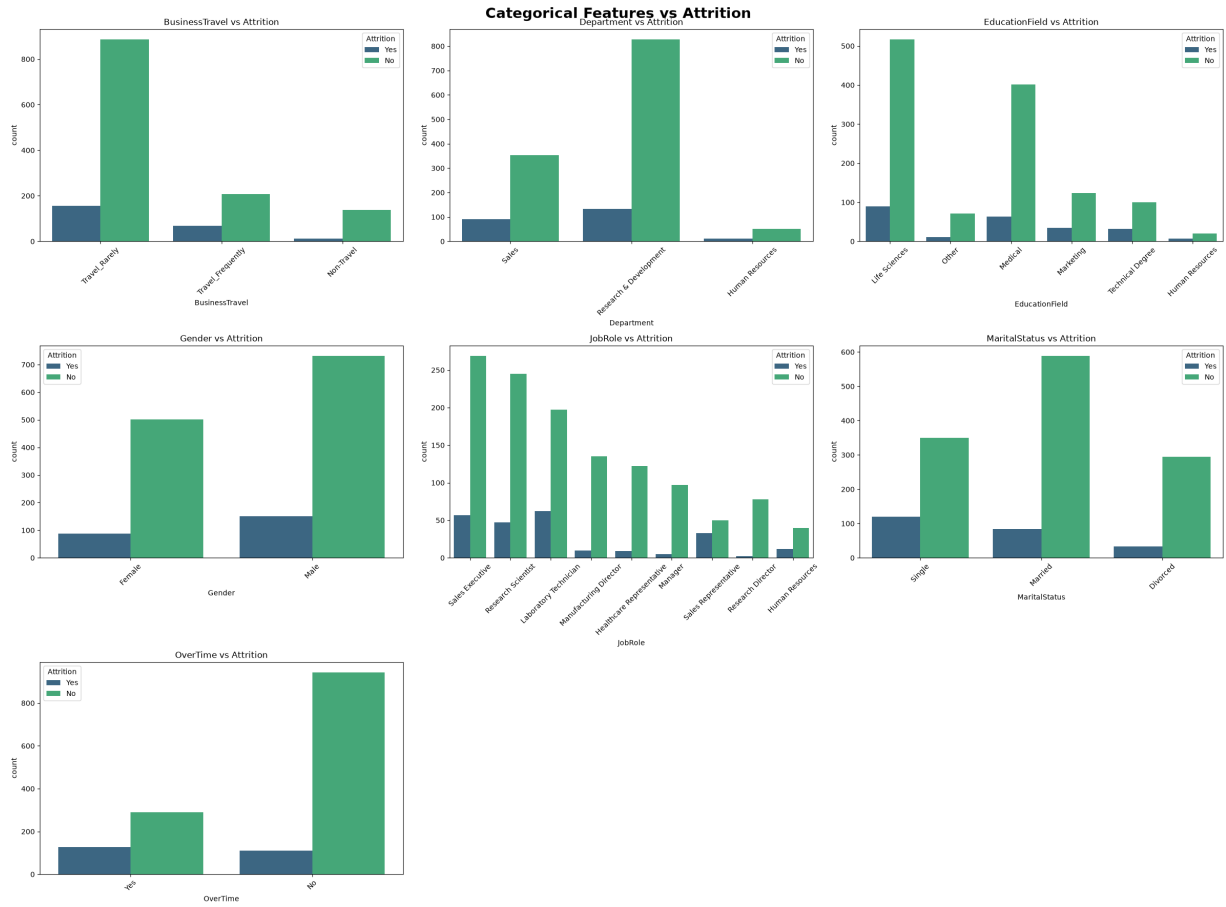
3.3 Numerical Feature Distributions

Distribution of Numerical Features



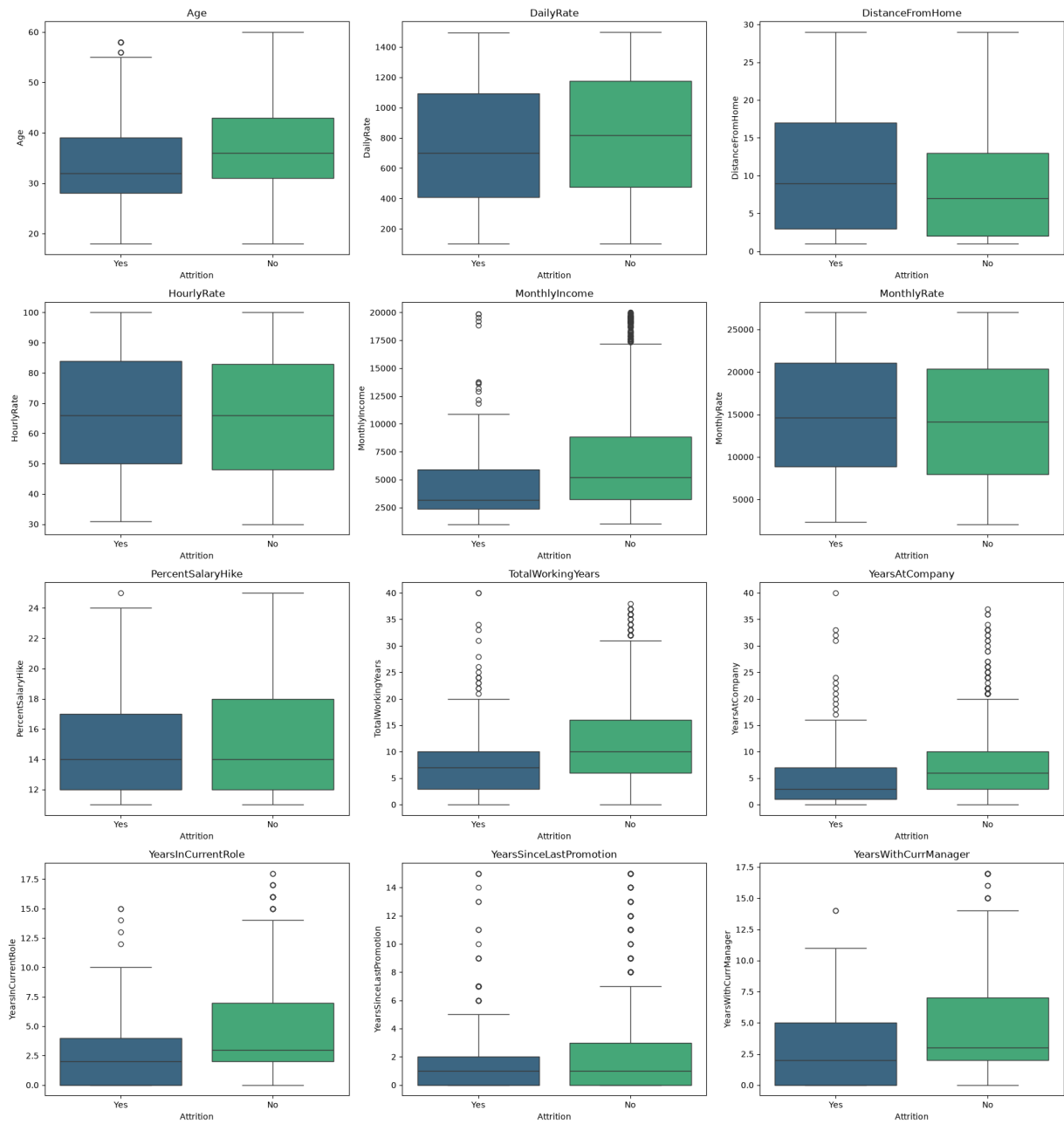
3.4 Continuous Features vs Attrition

Employees who leave tend to be younger, earn less, have fewer years at the company, and live further from home.



3.5 Ordinal Features vs Attrition

Continuous Numerical Features vs Attrition



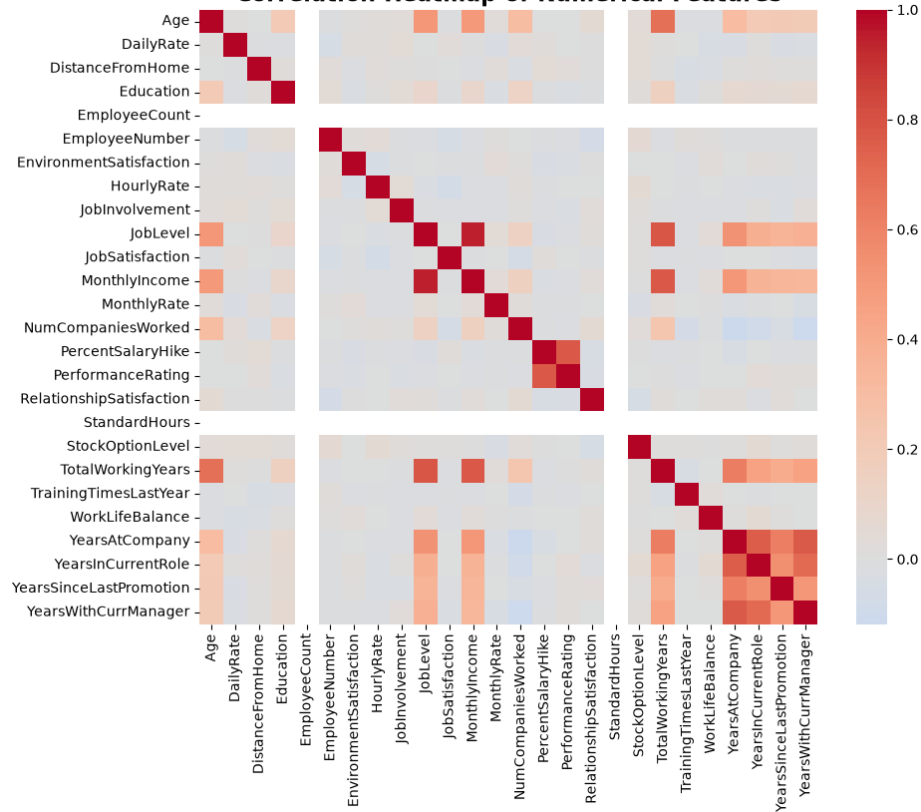
3.6 Categorical Features vs Attrition



3.7 Correlation Heatmap

Tenure variables are moderately correlated. JobLevel and MonthlyIncome are strongly correlated.

Correlation Heatmap of Numerical Features



4. Data Preprocessing

4.1 Feature Encoding

Encoding	Features
Ordinal	BusinessTravel
Binary	Gender, OverTime
One-Hot	Department, EducationField, JobRole, MaritalStatus
Target	Attrition → No=0, Yes=1

4.2 Feature Scaling

StandardScaler applied post-encoding. Fit on training data only to prevent leakage.

4.3 Class Imbalance — SMOTE

SMOTE was applied only to the training set to generate synthetic minority class samples and balance the 84/16 split. Test set remains imbalanced to reflect real-world conditions.

5. Model Building

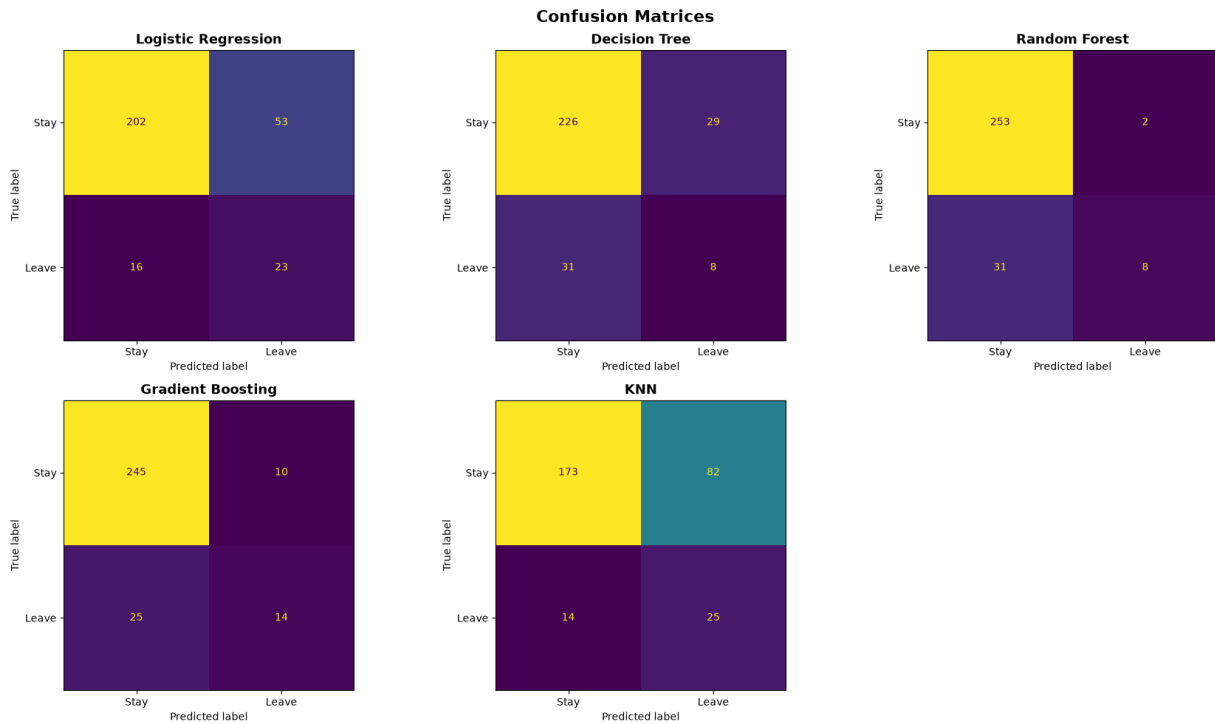
Five classifiers were trained on the SMOTE-resampled training data:

Model	Rationale
Logistic Regression	Linear baseline; fast and interpretable
Decision Tree	Non-linear; highly interpretable rules
Random Forest	Ensemble via bagging; reduces overfitting
Gradient Boosting	Sequential boosting; strong on tabular data
K-Nearest Neighbors	Distance-based; no distributional assumptions

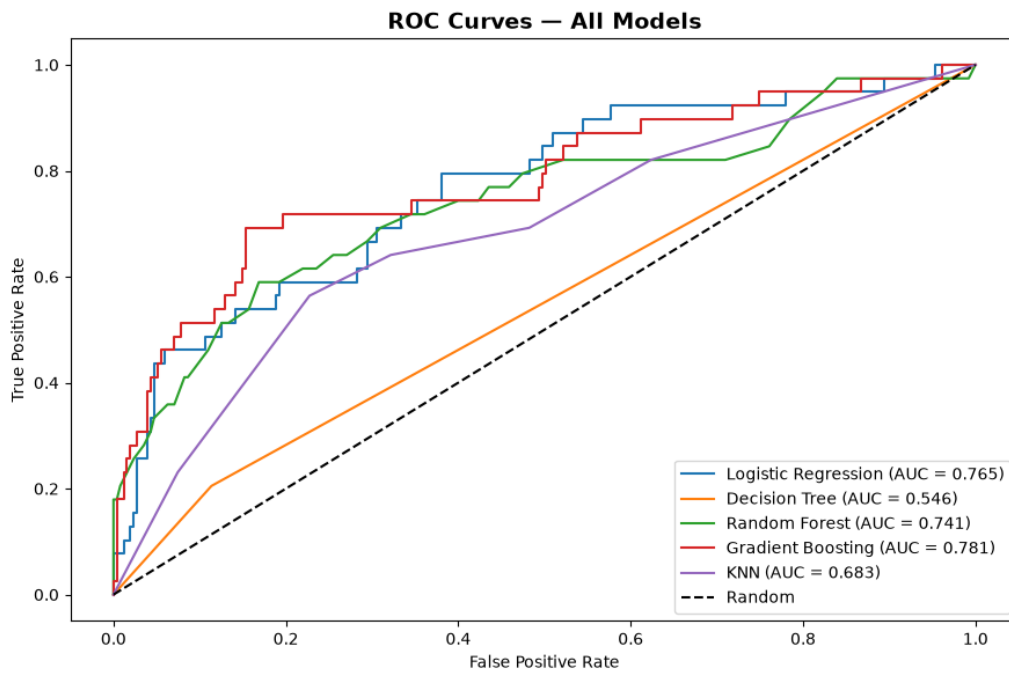
6. Model Evaluation

Models evaluated on original imbalanced test set (20% holdout). **Recall** is prioritised — missing an at-risk employee is more costly than a false alarm.

6.1 Confusion Matrices

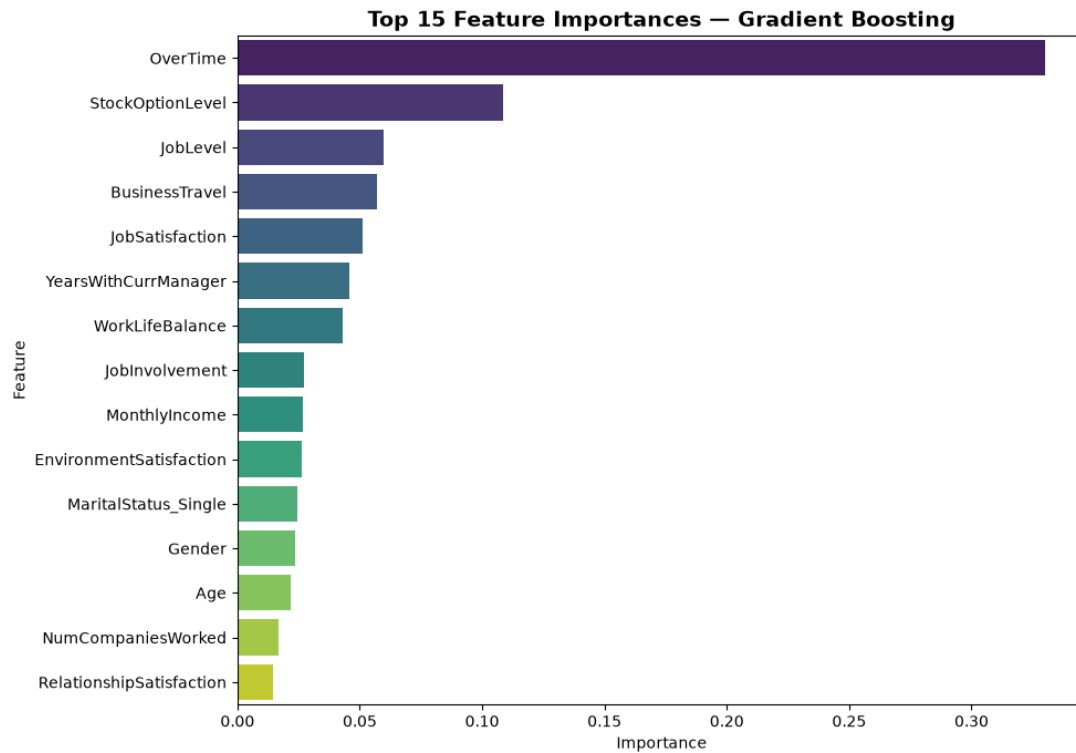


6.2 ROC Curves

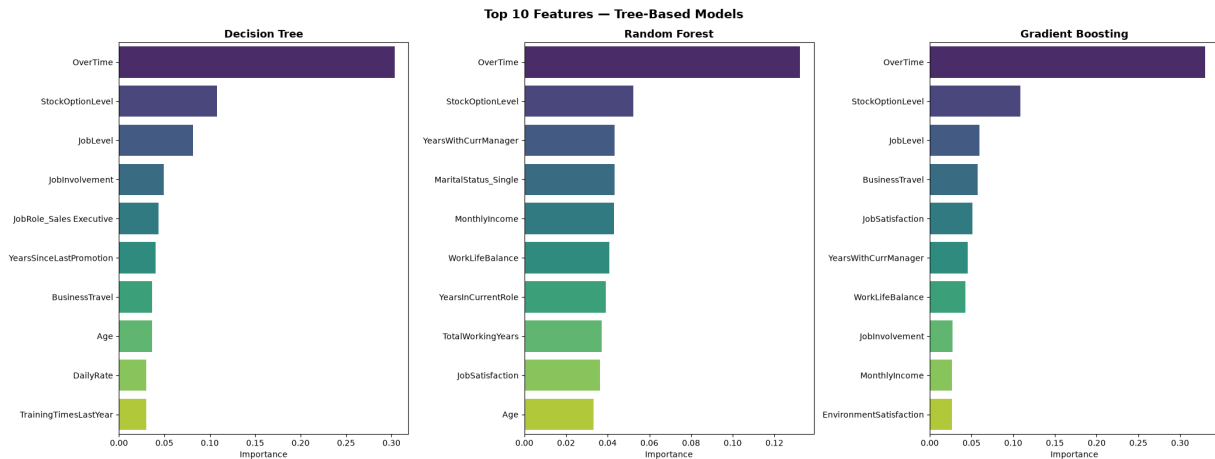


7. Feature Importance

Importances extracted from the best model (highest F1 Score). Consistent top predictors: **OverTime**, **MonthlyIncome**, **Age**, **TotalWorkingYears**, **JobLevel**.



Comparison — Tree-Based Models



8. Business Recommendations

Overtime Management

Monitor and cap overtime. Enforce work-life balance policies for high-risk roles.

Compensation Review

Benchmark salaries regularly. Provide clear, attainable promotion paths.

Early-Career Support

Target onboarding and mentorship for younger employees in their first 3 years.

Distance & Flexibility

Offer hybrid/remote options for employees with long commutes.

Satisfaction Monitoring

Run quarterly pulse surveys on environment, job, and work-life satisfaction with follow-up actions.

9. Conclusion

This project delivered an end-to-end attrition prediction pipeline — from EDA and preprocessing to model training, evaluation, and deployment-ready artefacts. The best model was selected by F1 Score and saved as model.pkl alongside the preprocessor and scaler.

Key finding: attrition is driven by workload (OverTime), compensation, career stage, and personal factors. Proactive HR monitoring of these signals enables early intervention before attrition occurs.